Partitives, Groups, and Wholes

David Nicolas[1] Jonathan D. Payton[2]

[1]Institut Jean Nicod - ENS, PSL, EHESS, CNRS

[2]Department of Philosophy - Bilkent University

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Two attractive metaphysical views

- **Identity Theory of Groups:**
  A group just *is* the individuals which make it up
- **Composition as Identity (CAI):**
  A whole is identical to its parts
  (collectively, not each individually)

Plan of the talk

1. Two problems for these views, drawn from plural partitives
2. A new semantic framework
3. How the framework dissolves the problems

Plural partitives

Plural partitive: an expression of the form [Det [of NP]], where the embedded NP is plural.

- "one of the guests"
- "two of several customers who complained"
- "some of them"

Plausible semantics (Ionin & Matushansky 2018):

- "of" indicates that any individuals satisfying the partitive are *among* those denoted by the plural NP.
- The determiner adds a specification, e.g. that there are n individuals.

Simple, plausible — and, we'll see, a source of trouble.

Problem 1 — The Identity Theory of Groups

Two teams: Ann & Bill;   Carrie & Daniel.
Suppose only Daniel is Québécois; the other three players are not.

(1) One of the players is Québécois.

(2) One of the teams is Québécois.

- One *player* is Québécois $\Rightarrow$ (1) is true.
- No *team* is Québécois
  (a team is Québécois only if all its members are) $\Rightarrow$ (2) is false.

But if the teams *are* the players, then "the teams" and "the players" co-denote.
The standard semantics predicts (1) and (2) have the *same* truth-value.
Contradiction.

Problem 2 — Composition as Identity

Suppose Tom and Jerry compose an object, *Genie*.

(3) Genie is one of them.

Case A: "them" is anaphoric on "Butch, Tom, and Jerry".
⇒ (3) is false.

Case B: "them" is anaphoric on "Butch and Genie".
⇒ (3) is true.

According to CAI, Genie is identical to Tom and Jerry.
So "them" denotes the same plurality in both cases and (3) should have the same truth-value.
Contradiction. (Yi 1999)

Our claim

Thesis

The arguments rest on a flawed semantics for plural partitives.
A better semantics dissolves both problems.

Key ideas:

- Plural partitives are **cover-sensitive**.
- In particular, a numeral must be interpreted with respect to the **basic cover** of the embedded NP.
- "is one of" is *not* the simple logical primitive standard plural logic assumes.

Covers

Inclusion, our one logical primitive: ‘$x \sqsubseteq y$’ reads ‘x is/are among y’.

A **cover** i is a partial, multivalued function. When i applies to a plurality a, its values are subpluralities of a. We say *i covers a* iff:

1. every value i assigns to a is included in a;
2. every individual included in a is included in some value i assigns to a.

Why covers? A familiar example

(4) Rodgers, Hart, and Hammerstein wrote musicals.

True if we carve the three men into:

- Rodgers & Hart, and
- Rodgers & Hammerstein,

and say of each pair that *they* wrote musicals.

(Gillon 1987)

But: not just *any* cover is available.

(5) The people lifted the piano. (distributes over people)

(6) The pairs of people lifted the piano. (distributes over pairs)

Fitting and basic covers

Fitting cover

i *fits* 'F' iff, when defined for x, it carves x into Fs and arbitrary combinations thereof.

Basic cover

$\star^F$ is the fitting cover which, when defined for x, carves x into *all and only* the Fs included in x.
(No Fs left out; no arbitrary combinations allowed.)

Inheritance: a nominal expression headed by 'F' inherits $\star^F$; so too a variable it restricts.

- "person", "people", "the people" $\rightsquigarrow$ $\star^{\text{Per}}$
- "pair of people", "the pairs of people" $\rightsquigarrow$ $\star^{\text{Pair}}$

Constraint: a predicate must be interpreted w.r.t. a cover that fits the head noun.

The semantics of numerals

Notation: '$y^i \precsim x^j$' reads "y are one of the values of j assigns to x".
(Cover-sensitive at its right argument: it depends on how x is carved up.)

Numeral schema

$$[\![n]\!] \;=\; \lambda i\, \lambda\phi\, \lambda x\, [\, \exists_n y\, (y^{\star^\phi} \precsim x^{\star^\phi})\ \&\ \phi(\star^\phi)(x)\,]$$

A numeral takes an intension ϕ to the property of being carved into exactly n subpluralities by the corresponding **basic cover** $\star^\phi$. For "one":

$$[\![\text{one}]\!] = \lambda i\, \lambda\phi\, \lambda x\, [\, \exists_1 y\, (y^{\star^\phi} \precsim x^{\star^\phi})\ \&\ \phi(\star^\phi)(x)\,] \qquad \text{i.e. being a single } \phi.$$

Key move: the numeral imports the basic cover of the NP into the truth-conditions.

Building "one of the teams"

Lexical entries, then composition (the partitive inherits $\star^T$ from "the teams"):

$$[\![\text{one}]\!] = \lambda i\, \lambda\phi\, \lambda x\, [\, \exists_1 y\, (y^{\star^\phi} \precsim x^{\star^\phi})\ \&\ \phi(\star^\phi)(x)\,]$$

$$[\![\text{of}]\!] = \lambda i\, \lambda x\, \lambda y\, [\, y^i \sqsubseteq x^i\,]$$

$$[\![\text{the teams}]\!] = \lambda i\, [t]$$

$$[\![\text{of the teams}]\!] = \lambda i\, \lambda x\, [\, x^i \sqsubseteq t^i\,]$$

$$[\![\text{one of the teams}]\!] = \lambda i \lambda x\, [\, \exists_1 y\, (y^{\star^T} \precsim x^{\star^T})\ \&\ x^{\star^T} \sqsubseteq t^{\star^T}\,]$$

In words: the property of being among the teams *and* being a (single) team.

Solving Problem 1 — Groups

Existential operator: $[\![\exists\exists]\!] = \lambda F_{<et>}\, \lambda G_{<et>}\, [\, \exists x\, (F(x)\ \&\ G(x))\,]$

$$[\![\text{One of the teams is Québécois}]\!]^{i,j} =$$

$$\exists x\, (\, \exists_1 y\, (y^{\star^T} \precsim x^{\star^T})\ \&\ x^{\star^T} \sqsubseteq t^{\star^T}\ \&\ \textsc{Québécois}(x^j)\,)$$

True iff some x:

- are a team (carved into one subplurality by $\star^T$),
- are among the teams,
- are Québécois.

In our scenario: no team is Québécois. $\Rightarrow$ false. ✓

For "one of the players": replace $\star^T$ with $\star^P$.
One player is Québécois. $\Rightarrow$ true. ✓

Both verdicts are correct — even though "the teams" and "the players" co-denote.

Solving Problem 2 — CAI

(3) Genie is one of them.

Truth-conditions:

$$\exists_1 y\,(y^\star \precsim g^\star)\ \&\ g^\star \sqsubseteq t^\star$$

What is $\star$? Inherited from the antecedent of “them”.

Basic covers (Compounds)

For a compound term $\alpha\beta$, $\star^{\alpha\beta}$’s values are all and only the values of $\star^\alpha$ and $\star^\beta$.

Solving Problem 2 — CAI (cont'd)

$$\exists_1 y\,(y^\star \precsim g^\star)\ \&\ g^\star \sqsubseteq t^\star$$

Case A: antecedent "Butch, Tom, and Jerry" $\rightsquigarrow$ $\star^{BTJ}$ carves t into Butch, Tom, Jerry.
Since Genie isn't among its values, $\star^{BTJ}$ does not carve g into a single cell.
$\Rightarrow$ "Genie is one of them" is false. ✓

Case B: antecedent "Butch and Genie" $\rightsquigarrow$ $\star^{BG}$ carves t into Butch and Genie.
Genie *is* a value, $\star^{BG}$ carves g into a single cell.
$\Rightarrow$ "Genie is one of them" is true. ✓

Both verdicts compatible with CAI. The two cases differ not in what "them" *denotes*, but in how that plurality is *carved up*.

What we've shown

- "of" translates as the cover-insensitive 'among' relation $\sqsubseteq$, as standardly.
- But its interaction with determiners makes plural partitives **cover-sensitive**.
- In particular, a numeral is interpreted relative to the **basic cover** of the NP.

Consequence for plural logic:
Standard plural logic takes "is one of" as a cover-insensitive primitive — which is what generates the problems. A better regimentation of "*a* is one of *b*":

$$\exists_1 x\,(x^\star \precsim a^\star)\ \&\ a^\star \sqsubseteq b^\star \qquad \text{(where } \star \text{ is the basic cover for '}b\text{')}$$

Unlike '$\sqsubseteq$', '$\precsim$' *is* cover-sensitive.

Consequence for metaphysics:
Two attractive views — the Identity Theory of Groups and CAI — escape objections drawn from plural partitives.

References

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Backup: beyond "one"

The schema generalizes. Suppose Ann and Daniel are both Québécois (one per team).

(7) Two of the players are Québécois. (true)

(8) Two of the teams are Québécois. (false)

Same shape of analysis; same kind of cover-sensitive divergence.

Numerically unspecific partitives also fall into place:

$$[\![\text{some}]\!] = \lambda i\,\lambda\phi\,\lambda x\,[\,\exists y\,(y^{\star^{\phi}} \precsim x^{\star^{\phi}})\ \&\ \phi(\star^{\phi})(x)\,]$$

$$[\![\text{all}]\!] = \lambda i\lambda\phi\lambda x[\forall y(\exists z(y^{\star^{\phi}} \precsim z^{\star^{\phi}}) \to y^{\star^{\phi}} \precsim x^{\star^{\phi}})\ \&\ \phi(\star^{\phi})(x)]$$

A general, attractive treatment of plural partitives — with metaphysical bite.

Backup: unpacking $[\![\text{all}]\!]$

$[\![\text{all}]\!] = \lambda i \lambda \phi \lambda x[\forall y(\exists z(y^{\star^\phi} \precsim z^{\star^\phi}) \rightarrow y^{\star^\phi} \precsim x^{\star^\phi}) \;\&\; \phi(\star^\phi)(x)]$

Antecedent $\exists z(y^{\star^\phi} \precsim z^{\star^\phi})$: y is a $\star^\phi$-cell of *something*.

Consequent: every such cell is a cell of x.

So 'All of the ϕs' requires x to exhaust the ϕ-cells and satisfy ϕ.

Backup: the arguments in plural logic

One-sorted plural language, neutral terms '$a, b, \ldots$'; basic predicate '$\sqsubseteq$' ('among').

- Identity: $a = b \equiv_{\mathrm{df}} a \sqsubseteq b \;\&\; b \sqsubseteq a$
- Individuality: $\mathsf{I}a \equiv_{\mathrm{df}} \forall x(x \sqsubseteq a \rightarrow x = a)$
- Singular inclusion: $a \leq b \equiv_{\mathrm{df}} \mathsf{I}a \;\&\; a \sqsubseteq b$

Standard reading: '$a \leq b$' = 'a is one of b'.

Regimentations:

"One of the players is Québécois." $\exists x(x \leq p \;\&\; \textsc{Québécois}(x))$

"One of the teams is Québécois." $\exists x(x \leq t \;\&\; \textsc{Québécois}(x))$

"Genie is one of them." $g \leq x$

Co-denoting terms yield equivalent claims (**contradiction**), because '$\leq$' is cover-insensitive.

Backup: compositional semantics

Modifies Heim & Kratzer (1998). Basic types: e (pluralities), t (truth-values), c (covers).

Intensional: $[\![\alpha]\!]$ is a function from one or more covers to an entity in the type hierarchy.

Functional application (FA)

If α is a branching node with daughters β, γ, and $[\![\beta]\!]^i$ is a function whose domain includes $[\![\gamma]\!]^j$, then

$$[\![\alpha]\!]^{i,j} = [\![\beta]\!]^i([\![\gamma]\!]^j).$$

Intensional functional application (IFA)

If α is a branching node with daughters β, γ, then for any i, if $[\![\beta]\!]^i$ is a function whose domain includes $\lambda j\,([\![\gamma]\!]^{j,k})$, then

$$[\![\alpha]\!]^{i,k} = [\![\beta]\!]^i(\lambda j\,([\![\gamma]\!]^{j,k})).$$

IFA lets a lexical item operate on the *intension* of its sister, not merely its extension.

Backup: full lexical entries

$$
\begin{aligned}
[\![\text{of}]\!] &= \lambda i\,\lambda x\,\lambda y\,[\,y^{i} \sqsubseteq x^{i}\,] \\
[\![\text{the teams}]\!] &= \lambda i\,[t] \\
[\![\text{Genie}]\!] &= \lambda j\,[g] \\
[\![\text{them}]\!] &= \lambda i\,[t] \\
[\![\text{(is) Québécois}]\!] &= \lambda i\,\lambda x\,[\,\textsc{Québécois}(x^{i})\,] \\
[\![\text{n}]\!] &= \lambda i\,\lambda\phi\,\lambda x\,[\,\exists_{n} y\,(y^{\star^{\phi}} \precsim x^{\star^{\phi}})\;\&\;\phi(\star^{\phi})(x)\,] \\
[\![\exists\exists]\!] &= \lambda F_{<\mathrm{et}>}\,\lambda G_{<\mathrm{et}>}\,[\,\exists x\,(F(x)\;\&\;G(x))\,]
\end{aligned}
$$

Backup: deriving "of the teams"

By FA:

$$
\begin{aligned}
[\![\text{of the teams}]\!]^{i,j} &= [\![\text{of}]\!]^{i}([\![\text{the teams}]\!]^{j}) \\
&= \lambda x\, \lambda y\, [y^{i} \sqsubseteq x^{i}](t) = \lambda y\, [\, y^{i} \sqsubseteq t^{i}\,]
\end{aligned}
$$

Abstracting on i:

$$[\![\text{of the teams}]\!] = \lambda i\, \lambda x\, [\, x^{i} \sqsubseteq t^{i}\,]$$

Condition (i): being among the teams. Cover-insensitive (i inert), but the intension inherits $\star^{T}$ — which (ii) exploits.

Backup: deriving "one of the teams"

Numeral (basic cover $\star^{\phi}$ contributed by the argument):

$$[\![\text{one}]\!] = \lambda i\, \lambda\phi\, \lambda x\, [\, \exists_1 y\, (y^{\star^{\phi}} \precsim x^{\star^{\phi}})\ \&\ \phi(\star^{\phi})(x)\,]$$

By IFA, feeding it $[\![\text{of the teams}]\!]$ (so $\star^{\phi} = \star^{T}$):

$$\begin{aligned}
[\![\text{one of the teams}]\!]^{i} &= [\![\text{one}]\!]^{i}([\![\text{of the teams}]\!]) \\
&= \lambda x\, [\, \exists_1 y\, (y^{\star^{T}} \precsim x^{\star^{T}})\ \&\ \lambda j \lambda z [z^{j} \sqsubseteq t^{j}](\star^{T})(x)\,] \\
&= \lambda x\, [\, \exists_1 y\, (y^{\star^{T}} \precsim x^{\star^{T}})\ \&\ x^{\star^{T}} \sqsubseteq t^{\star^{T}}\,]
\end{aligned}$$

Condition (ii): $\star^{T}$ carves x into exactly one cell — i.e. x are *a team.*

Backup: deriving the whole sentence

Existential operator and predicate:

$$[\![\exists\exists]\!] = \lambda F\, \lambda G\, [\, \exists x\, (F(x)\ \&\ G(x))\,] \qquad [\![\text{(is) Québécois}]\!] = \lambda i\, \lambda x\, [\, \textsc{Québécois}(x^i)\,]$$

By FA:

$$\begin{aligned}[\![\text{One of the teams is Québécois}]\!]^{i,j} &= \exists x\, ([\![\text{one of the teams}]\!]^{i}(x)\ \&\ \textsc{Québécois}(x^j)) \\ &= \exists x\, (\, \exists_1 y\, (y^{\star^T} \precsim x^{\star^T})\ \&\ x^{\star^T} \sqsubseteq t^{\star^T}\ \&\ \textsc{Québécois}(x^j)\,)\end{aligned}$$

True iff some x: are a team, are among the teams, are Québécois.

Free j constrained: ‘x’ inherits $\star^T$, so distribution stops at teams.

Backup: distributive predicates

In $\mathcal{CPL}$:

$$F^D(a^i) \;\equiv_{\text{df}}\; \forall x\,(x^i \precsim a^i \rightarrow Fx^i)$$

So whether F^D is true of a depends on the choice of cover i.

The basic cover of "the people" is $\star^{\text{Per}}$, the basic cover of "the pairs" is $\star^{\text{Pair}}$. This explains why "The people lifted the piano" and "The pairs of people lifted the piano" have distinct (purely) distributive readings.

Backup: bands and teams

(9) The bands and the teams met.

A reading on which “met” distributes over each band and over each team.

By the compound rule:

$$\star^{BaTe}(e) = \text{all and only the bands and teams among } e$$

This independently motivates the compound-cover principle used in the Genie case.

Backup: covers vs. higher-level plurals

(10) The red socks, the wool socks, and the perforated socks overlap.

True iff some sock is red, wool, and perforated.

HLP remedy: subject denotes a plurality *of* three sock-pluralities; contested.

Our remedy: one plurality of socks, ‘overlap’ evaluated under a cover into the red, wool, perforated subpluralities. Work done by a plurality *of* pluralities is done by a single plurality *under* a cover.